

Performance Testing

Date	02-07-2026
Team ID	
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust, Postman
Type of Testing	Load Testing
Target Module / API	Credit Prediction API Endpoint
Test Environment	Local
Test Date	03-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline load test	10	60	No errors, stable response
2	Moderate load test	50	60	Response time < 2s
3	Peak load test	100	60	Performance remains stable

4				
---	--	--	--	--



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.8 seconds	Pass	Within limits
2	Response Time (Max)	< 5 seconds	1.5 seconds	Pass	stable
3	Throughput (Req/sec)		12 Req/sec	Pass	Good
4	Error Rate	< 1%	0%	Pass	No Errors
5	CPU Utilization	< 80%	25%	Pass	Efficient
6	Memory Utilization	< 80%	30%	Pass	Stable

Step 4: Observations & A

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

```
model.ipynb X
C:\Users\N Indhu > OneDrive > Desktop > credit_project > Project Development Phase > model.ipynb > # 1. Data Analysis Libraries
Generate + Code + Markdown | Run All X Clear All Outputs | Outline ...
25% 0.000000
...
Approval_Status 0
dtype: int64

>>> Epic 4: Comparative Machine Learning Modeling (All 4 Algorithms) <<<
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
c:\Users\N Indhu\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\linear_model\_logistic.py:599: ConvergenceWarning
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=1000).
You might also want to scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear\_model.html#logistic-regression
n_iter_1 = _check_optimize_result(
Logistic Regression Model Accuracy: 97.00%
Decision Tree Model Accuracy: 95.00%
Random Forest Model Accuracy: 95.00%
XGBoost Model Accuracy: 96.00%

👉 Best Performing Model Selected for Deployment: Logistic Regression (97.00%)
>>> Success: Saved 'best_credit_model.pkl' into the project folder local directory! <<<
```

```
model.ipynb X
C:\Users\N Indhu > OneDrive > Desktop > credit_project > Project Development Phase > model.ipynb > # 1. Data A
Generate + Code + Markdown | Run All X Clear All Outputs | Outline ...
No 235
Name: count, dtype: int64

--- Descriptive Statistics Summary ---

```

	Applicant_ID	Age	Income_Total	Employment_Years
count	500.000000	500.000000	500.000000	500.000000
mean	1250.500000	43.116000	85603.976000	9.952200
std	144.481833	12.733217	38162.832321	5.880392
min	1001.000000	21.000000	20526.000000	0.000000
25%	1125.750000	32.000000	51582.500000	5.200000
50%	1250.500000	44.000000	85548.000000	9.750000
75%	1375.250000	53.000000	118821.750000	15.200000
max	1500.000000	64.000000	149839.000000	20.000000

```

Approval_Status
count      500.000000
mean         0.500000
std          0.500501
min          0.000000
25%          0.000000
...
Approval_Status 0
dtype: int64

>>> Epic 4: Comparative Machine Learning Modeling (All 4 Algorithms) <<<
```